Auditing Causal-Discovery Systems: An Open Protocol and a First Audit of the Field

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Abstract

Causal-discovery systems are now judged by a single benchmark score, and language-model agents such as Causal-Copilot run the whole analysis on their own. Neither kind of report says where an answer comes from: the numbers, the variable names, the file layout, or the system's own rules. We propose an open protocol for auditing causal-discovery systems as black boxes or, where code is open, stage by stage. It holds the numbers fixed while names and column order change, traces every intermediate graph, types each failure, and reports scores only within a stated regime and against a reference of stated reliability. Under it, the first audit of the field finds the same pattern in benchmarks, in agents, and inside one agent. Benchmark scores reward layout: on CausalArena, an undecided-pair rule that follows the released column order adds 0.221 directed F1 to FCI. Agents read names: swapping the names costs 3 of 4 entered systems at least 0.10 directed F1. And an agent's own steps can undo its data: Causal-Copilot's review lowers the adjacency F1 of its algorithm's graph from 0.584 to 0.321, and every one of the 8 runs that reached its cycle step crashed. The protocol, the recorded runs and the scripts behind every number are public.

Code: https://github.com/uzairgheewala/causal-copilot-q1

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1 Introduction

Many decisions turn on a causal question: if we change this, what else moves? When experiments are impossible or expensive, the structure is learned from observational data, and every discovery algorithm brings assumptions that an expert must check against the problem (Spirtes et al. 2000). Language-model agents now promise to automate that expert. Causal-Copilot inspects the data, chooses and configures an algorithm, has a language model review the graph, and writes the report (Wang et al. 2025); several other systems let a model supply constraints, a causal order or edits to a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Vashishtha et al. 2025).

How far can such a system's claims be trusted? Today the answer is a score on a benchmark. A score cannot say where the answer came from, and there are known ways to earn one without the data: simulated benchmarks leak the causal order through the scale of the variables (Reisach et al. 2021), the column order of the input file changes what an algorithm learns (Kitson and Constantinou 2024), and language models reconstruct well-known graphs from the dataset and variable names alone (Srivastava et al. 2025). A recent position paper asks for audit trails that keep data, assumptions and agent recommendations apart (Zheng et al. 2026). What is missing is a routine way to check a deployed system for these failures, with controls, and to compare systems on the result.

We propose such a protocol. Its target is a continuous-integration check for causal claims: point it at a system through a small adapter and receive an audit report that says what the system gets right, where its answers come from, how its errors behave under controlled change, and whether its confidence means anything. The protocol has levels, each usable alone. L0 scores the final graph against a known one. L1 runs metamorphic controls: the same numbers under real, neutral and swapped names, with the columns shuffled and the file paths made opaque, so a change in the answer can only come from the names or the layout. L2 traces a system stage by stage where its code is open, recording which step adds, deletes or reverses each edge and which model call caused it. L3 (claim provenance) and L4 (adversarial search) are designed and come later. Every finding is typed from a closed list (name reliance, layout reliance, deletion of links the data support, crash, cost outlier, and so on), carries the control that shows it and an interval, and is judged against a threshold written and hashed before the run. A score is compared only within its regime (data only, or with interventions, or with declared knowledge), and a reference graph carries a reliability tier: exact when the data were simulated from it, contested when published versions disagree, provenance-only when nothing can be compared against it, and mismatched when it differs from the version a benchmark uses.

The first audit under the protocol covers 5 language-model systems and the CausalArena and CausalBench benchmarks, with Causal-Copilot as a worked case traced stage by stage. It finds the same pattern in benchmarks, in agents, and inside one agent.

- **Benchmarks reward shortcuts.** CausalArena's released files list the variables in causal order, and its scoring forces a direction on every undecided pair, broken by column index; FCI gains 0.221 directed F1 on the semantic family from that rule alone, and the published PC row (0.333) rises to 0.463 with an order-free collider

rule on shuffled columns. On CausalBench, the main metric rewards ranking total effects: MeanDifference reaches 0.90–0.95 of the ceiling set by the test split's own knockdown effects, and the second metric, the false omission rate, does not separate it from random networks.

- **Agents read names, not data.** With the numbers fixed, swapping the names costs 3 of the 4 systems that entered the pre-registered test (at least 5 paired scenarios each) at least 0.10 directed F1 (causaliq-knowledge 0.251, Causal-Copilot 0.167, MATMCD 0.122). The system that resists, the pipeline of Vashishtha et al. (2025) (0.020), is the one whose data decide the graph. Inside Causal-Copilot, 81 of the 101 distinct edges the model set in swapped-name runs follow the labels rather than the data, against 20 that follow the data.
- **An agent's own steps can undo its data.** MATMCD's final graph has lower committed-claim precision than its own data-only graph (0.6855 against 0.7199). Causal-Copilot's review step lowers the adjacency F1 of its algorithm's graph from 0.584 (all 30 tables) to 0.321 (the 22 runs that finished), and all 8 runs that reached its cycle step crashed there (a missing key in the model's reply, 8 of 8).

None of the vendors we reviewed publishes how often its causal claims are right, or how often the system should say "I don't know". The audit therefore also sets a bar: a system's claims should come with their source and with a confidence that survives these controls.

Contributions. (i) An open audit protocol with typed findings, regimes, truth tiers, pre-registered thresholds and a reproducible certificate. (ii) The first audit of the field under it: name reliance across 5 systems, and layout and metric shortcuts in CausalArena and CausalBench. (iii) A stage-level account of Causal-Copilot from recorded runs, with every model call and intermediate graph kept. (iv) The code, the recorded runs and the scripts that recompute every number in this report. A companion project uses the audit's findings to rebuild Causal-Copilot's review step; it is audited through the same interface and is not reported here.

This checkpoint covers the introduction, the prior work and the data. The protocol's full specification, the audit package and the results tables belong to the final report.

2 Prior work

2.1 How causal discovery is evaluated, and what benchmarks reward

Constraint-based algorithms such as PC and FCI recover causal structure from conditional-independence tests, and only up to an equivalence class under assumptions such as faithfulness (Spirtes et al. 2000). Score-based search such as GES works over equivalence classes and, with a consistent score and a distribution that has a DAG perfect map, identifies the structure in the large-sample limit (Chickering 2002). Benchmarks with known graphs, on which these methods are compared, have been shown to carry shortcuts (Reisach et al. 2021; Kitson and Constantinou 2024). In common simulated settings the variance of the variables grows along the causal order, so sorting by variance matches continuous-

optimisation methods (Reisach et al. 2021), and a scale-invariant version of the same artefact is also high in common settings and can be exploited (Reisach et al. 2023). The order of the columns in the input data alone changes the graph that hill-climbing search learns, typically more than a change in sample size does (Kitson and Constantinou 2024). Real-data evaluation leans on very few datasets, above all the Sachs protein-signalling data, whose reference graph is not definitive (Brouillard et al. 2025; Sachs et al. 2005). CausalArena collects synthetic, semantic and formula-grounded models with a public leaderboard (Li et al. 2026), and CausalBench scores gene-network inference on held-out perturbation data (Chevalley et al. 2025; Replogle et al. 2022). These works audit benchmarks or algorithms one artefact at a time; none applies a common set of controls to the systems that are now deployed.

2.2 Language models as a source of causal knowledge

Language models answer pairwise causal questions about named variables well, and part of that skill is memorised (Kiciman et al. 2023); they may repeat causal statements without performing causal inference (Zečević et al. 2023). When a model is treated as an imperfect expert, the guarantee that the true graph is kept holds only under an assumed error model, and in the authors' own case study the model was poorly calibrated and the true graph was excluded on one benchmark (Long et al. 2023). Asking for a causal order instead of individual edges avoids confusing direct with indirect effects (Vashishtha et al. 2025). Controls for memorisation now exist: neutral aliases in the model's queries (Kitson and Constantinou 2026), graphs published after a model's training cut-off (Srivastava et al. 2025), random graphs carrying real concept names (Li and Russo 2026), and names moved onto the wrong columns with the data unchanged, under which models' accuracy collapses (Dong et al. 2025). The last is the control we use; we add the layout controls and apply both to deployed agents and to benchmarks.

2.3 Causal agents and how they are evaluated

Causal-Copilot automates the whole analysis and states that the model's recommendations are soft suggestions that may not override the high-confidence decisions of its bootstrap procedure (Wang et al. 2025). ALCM, MATMCD and Causal Modelling Agents let a model refine or constrain a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Abdulaal et al. 2024), and open libraries expose model-based causal suggestions (CausalIQ 2026; PyWhy 2026). These systems are mostly evaluated on well-known named benchmarks against the plain algorithm. The only independent evaluation of Causal-Copilot we found scores it as a baseline and, when necessary, steered it towards a tractable algorithm because it often preferred intractable ones (Tagliapietra et al. 2026). Repeated runs of the same agent can disagree, so single-run comparisons can misrank systems (Mehta 2026). A position paper argues that agents should not supply edges or causal conclusions at all and asks for audit trails (Zheng et al. 2026). We found no stage-by-stage evaluation of a deployed causal-discovery agent under name and layout controls.

2.4 How much an edge can be trusted

Bootstrap resampling gives confidence measures for features of a learned network (Friedman et al. 1999), but the resulting edge frequencies can be miscalibrated (Kummerfeld and Rix 2019; Banerjee et al. 2025). Language models asked about direct edges give high stated confidence to most of their false-positive edges; agreement across prompts is better calibrated, though not significantly so (Kumar et al. 2026). An audit therefore has to test whether a system's confidence means anything, not assume it.

2.5 Auditing practice the protocol borrows

Metamorphic testing checks a program that has no oracle for each output by checking relations between the outputs of related inputs (Chen et al. 1998; Segura et al. 2016); renaming or reordering columns while holding the numbers fixed is such a relation. Behavioural testing of language-processing models crosses capabilities with test types that include invariance tests, and finds failures that held-out accuracy hides (Ribeiro et al. 2020). Model cards and datasheets document a model's evaluation and a dataset's origin (Mitchell et al. 2019; Gebru et al. 2021), internal algorithmic audits define staged audit artefacts (Raji et al. 2020), and PROV-DM records which agent and activity produced a piece of data (Moreau and Missier 2013). Our certificate combines these ideas for causal claims: system identity, battery hash, every number with its source, and the command that reruns it.

3 Data

The audit draws on a suite built from CausalArena's semantic scenarios, Sachs and 4 other named datasets that ship with Causal-Copilot, and CausalBench's perturbation data. Table 1 gives their shapes, the reliability tier of each reference graph and the licences. All shapes are read from the suite manifests (Appendix A.1); the suites were frozen before any run and are checked by hash.

CausalArena semantic suite. CausalArena releases structural causal models whose variables carry names from a written scenario, with observational exports (Li et al. 2026). We take the 10 semantic scenarios of our test split, the first replicate of the observational-only release, 1,000 rows each. Each scenario is written 3 times with byte-identical numbers: under its real names, under neutral names, and under its own names moved onto the wrong columns by a derangement that minimises pairs adjacent under both readings (30 tables). Because the released column order is a causal order, the columns are shuffled with a fixed seed, and the file paths are opaque codes. The reference is the graph the data were simulated from, so scores are accuracy. The layout audit of the public leaderboard uses CausalArena's public files instead (250 semantic, 250 formula and 100 synthetic). The dataset is released under CC BY 4.0 on Hugging Face (`LAMDA-Tabular/CausalArena`).

Table 1: Data used in the audit. Rows and variables of the files as given to the systems; for CausalBench, cells and perturbed genes after CausalBench's own filtering. Tier: reliability of the reference graph (Section 1).

Group	Dataset	Rows	Variables	Ref. edges	Tier	Licence
CausalArena	10 semantic scenarios	1,000	20–25	35–58	exact	CC BY 4.0
Named	Sachs	853	11	19	contested	none stated
	Abalone	4,177	8	7	provenance	CC BY 4.0
	Auto MPG	392	5	5	mismatched	CC BY 4.0
	Concrete strength	1,030	9	8	provenance	CC BY 4.0
	DWD climate	349	5	4	provenance	CC BY 4.0
CausalBench	K562	162,751	622	–	held-out interventions	CC BY 4.0
	RPE1	162,733	383	–	held-out interventions	CC BY 4.0

Named datasets. Causal-Copilot ships Sachs and 4 other tables with reference graphs, and we use its copies (Wang et al. 2025). Sachs (Sachs et al. 2005) has 4 arms (the original, a copy under another path, neutral names, and moved names). Its reference is contested: the repository's graph has 19 edges, the bnlearn version 17, sharing 16 adjacencies with 1 reversed, and CausalArena uses a version with 20 edges, so a score on Sachs is agreement with one version, not accuracy. Abalone (Nash et al. 1994), Auto MPG (Quinlan 1993) and Concrete compressive strength (Yeh 1998) come from the UCI repository under CC BY 4.0, and the climate table matches the Deutscher Wetterdienst pairs of the Tübingen cause-effect benchmark (Mooij et al. 2016); DWD's open data are released under CC BY 4.0. Each has 3 arms (real, neutral, swapped). Their reference graphs are expert graphs with nothing to compare against (provenance tier), except Auto MPG, whose reference differs from CausalArena's version (4 variables, 3 edges, 398 rows), so its scores must not be set beside CausalArena's row. No licence is stated for the Sachs data.

CausalBench. CausalBench uses the CRISPRi Perturb-seq screens of Replogle et al. (2022) in the K562 and RPE1 cell lines and scores a predicted network on held-out interventions by a mean Wasserstein distance and a false omission rate (Chevalley et al. 2025). After CausalBench's own filtering, which we reproduced, the K562 data have 162,751 cells (10,691 unperturbed controls) and 622 perturbed genes, and RPE1 162,733 cells (11,485 controls) and 383 genes. The screens are released under CC BY 4.0 and the benchmark code under Apache 2.0. Our audit of this benchmark is cited from its result file, not re-run: all 4 published MeanDifference rows were reproduced; MeanDifference reaches 0.90–0.95 of the ceiling set by ranking pairs by the knockdown effects measured on the test split itself, so the main metric rewards ranking total effects; and its false omission rate stays within the pre-registered margin of random networks in 3 of the 4 cells, the remaining one being undefined.

Recorded runs. A recorded run keeps the system's identity (code hash checked before and after the run, model snapshot), every model request and reply, the graph after every stage, and the refinement's intermediate steps. Causal-Copilot was run unmodified (upstream commit `a31aa7c`) with gpt-4o-2024-08-06: 30 runs on the semantic suite in the name-arm study, and a main-control series of 30 runs on the same tables (API cost $12.57). The other systems were run through adapters on the same suite.

Limits of the data. The semantic scenarios are simulated, so their exactness comes with simulation's artefacts; each table and arm was run once; one model family was used; and the named datasets' references are expert graphs. The protocol treats each of these as a declared regime or tier rather than hiding it.

Code

The public repository (`https://github.com/uzairgheewala/causal-copilot-q1`) recomputes the Causal-Copilot stage comparison from the recorded runs with one command and no API key, and documents the controls and metrics. The numbers in this report are read from result files of the research repository (`https://github.com/uzairgheewala/cc-research`) at commit `cf536c12df7f`; Appendix A.1 lists each with its file and JSON pointer.

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Appendices

A.1 Where every number comes from A1

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Generated by `docs/paper/q1/build_numbers.py` from `numbers.json`. Paths are relative to the root of cc-research; a pointer is a JSON pointer, where `*` takes every child and `[k=v]` the first list item with that field. A flag after a source marks a file that was not unchanged in git when the report was built.

Key	Value	Source file and JSON pointer (cc-research)
`cc.arms`	30	`analysis/results/fork/bld06_live_compare.json /arms`
`cc.completed`	22	`analysis/results/fork/bld06_live_compare.json /completed/main`
`cc.s0-adj`	0.584	`analysis/results/fork/bld06_live_compare.json /means/main/S0_adj_f1`
`cc.final-adj`	0.321	`analysis/results/fork/bld06_live_compare.json /means/main/final_adj_f1`
`cc.usd`	$12.57	`analysis/results/fork/bld06_live_compare.json /usd/main`
`cc.cyc.entered`	8	`analysis/results/fork/bld06_cycles.json /arms/main/runs_entering_cycle_step`
`cc.cyc.keyerr`	8	`analysis/results/fork/bld06_cycles.json /arms/main/cycle_step_keyerrors`
`bnd08.runs`	30	len of: `analysis/results/stability_s2/bnd08_summary.json /runs`
`bnd08.m3.labels`	81	`analysis/results/stability_s2/bnd08_summary.json /M3/labels_only`
`bnd08.m3.data`	20	`analysis/results/stability_s2/bnd08_summary.json /M3/data_only`
`bnd08.m3.n`	101	`analysis/results/stability_s2/bnd08_summary.json /M3/n`
`bnd08.model`	gpt-4o-2024-08-06	`analysis/results/stability_s2/bnd08_summary.json /snapshots/0`
`ab06.agents`	4	len of: `analysis/results/ab06/ab06.json /agents`
`ab06.cc-entered`	1	count_true of: `analysis/results/ab06/ab06.json /causal_copilot/*/counted_in_P9`

Key	Value	Source file and JSON pointer (cc-research)
ab06.systems	5	add of: key ab06.agents; key ab06.cc-entered
ab06.p9.entered	4	len of: analysis/results/ab06/ab06.json /P9/systems_entered
ab06.p9.losing	3	len of: analysis/results/ab06/ab06.json /P9/losing_ge_0.10
ab06.p9.min-scen	5	analysis/results/ab06/ab06.json /P9/min_paired_scenarios
ab06.p9.threshold	0.10	plans/ab06_prereg.json /predictions/P9/supported (pattern lose ⩾ ([0-9.]+))
ab06.causaliq.drop	0.251	analysis/results/ab06/ab06.json /agents/causaliq/name_reliance_real_minus_swapped/mean
ab06.cc.drop	0.167	analysis/results/ab06/ab06.json /causal_copilot/causal_copilot_gpt4o/name_reliance_real_minus_swapped/mean
ab06.matmcd.drop	0.122	analysis/results/ab06/ab06.json /agents/matmcd/name_reliance_real_minus_swapped/mean
ab06.vashishtha.drop	0.020	analysis/results/ab06/ab06.json /agents/vashishtha/name_reliance_real_minus_swapped/mean
ab06.matmcd.dv1	0.6855	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_final
ab06.matmcd.dv1-data	0.7199	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_data_only
ab01.pc.published	0.333	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, default rule);family=semantic]/published
ab01.pcmaxp.controlled	0.463	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, maxP);family=semantic]/ours_shuffled_expectation
ab01.fci.authors	0.578	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_authors_rule
ab01.fci.expect	0.357	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_expectation

Key	Value	Source file and JSON pointer (cc-research)
ab01.fci.gain	0.221	sub of: key ab01.fci.authors; key ab01.fci.expect
ab01.files.semantic	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=semantic]/files
ab01.files.formula	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=formula]/files
ab01.files.synthetic	100	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=synthetic]/files
ben13.reproduced	4	analysis/results/ben13/summary.json / reproduced_rows
ben13.cells	4	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/placement_share_wasserstein
ben13.for.random-level	3	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/?placement_share_for_note
ben13.echo.min	0.90	min of: analysis/results/ben13/summary.json / predictions/P3/values/*
ben13.echo.max	0.95	max of: analysis/results/ben13/summary.json / predictions/P3/values/*
s2.families	10	len of: datasets/suite_q1s2/MAPPING.json / families
s2.tables	30	len of: datasets/suite_q1s2/MAPPING.json /arms
s2.arms-per	3	div of: key s2.tables; key s2.families
s2.rows	1,000	max of: datasets/suite_q1s2/MAPPING.json / families/*/rows
s2.vars.min	20	min of: datasets/suite_q1s2/MAPPING.json / families/*/variables
s2.vars.max	25	max of: datasets/suite_q1s2/MAPPING.json / families/*/variables
s2.edges.min	35	min of: datasets/suite_q1s2/MAPPING.json / families/*/edges
s2.edges.max	58	max of: datasets/suite_q1s2/MAPPING.json / families/*/edges
s1.sachs.rows	853	datasets/MAPPING.json /source/rows
s1.sachs.vars	11	len of: datasets/MAPPING.json /source/columns
s1.sachs.edges	19	len of: datasets/MAPPING.json /source/reference_edges
s1.sachs.arms	4	len of: datasets/MAPPING.json /arms

Key	Value	Source file and JSON pointer (cc-research)
s1.sachs.bnlearn	17	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/edges_b
s1.sachs.shared	16	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/shared_adjacencies
s1.sachs.reversed	1	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/reversed
s1.sachs.arena	20	analysis/results/reference_reliability/reliability.json /families/sachs/causalarena_version/edges
s1.abalone.rows	4,177	datasets/suite_b1/MAPPING.json /families/abalone/rows
s1.abalone.vars	8	len of: datasets/suite_b1/MAPPING.json /families/abalone/columns
s1.abalone.edges	7	len of: datasets/suite_b1/MAPPING.json /families/abalone/reference_edges
s1.autompg.rows	392	datasets/suite_b1/MAPPING.json /families/autompg/rows
s1.autompg.vars	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/columns
s1.autompg.edges	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/reference_edges
s1.concrete.rows	1,030	datasets/suite_b1/MAPPING.json /families/concrete/rows
s1.concrete.vars	9	len of: datasets/suite_b1/MAPPING.json /families/concrete/columns
s1.concrete.edges	8	len of: datasets/suite_b1/MAPPING.json /families/concrete/reference_edges
s1.dwd.rows	349	datasets/suite_b1/MAPPING.json /families/dwd/rows
s1.dwd.vars	5	len of: datasets/suite_b1/MAPPING.json /families/dwd/columns
s1.dwd.edges	4	len of: datasets/suite_b1/MAPPING.json /families/dwd/reference_edges
s1.autompg.arena.vars	4	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/0
s1.autompg.arena.edges	3	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/1

Key	Value	Source file and JSON pointer (cc-research)
s1.autompg.arena.rows	398	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/2
s1.named.tables	12	len of: datasets/suite_b1/MAPPING.json /arms
s1.named	4	len of: datasets/suite_b1/MAPPING.json /families
s1.named.arms-per	3	div of: key s1.named.tables; key s1.named
cb.k562.cells	162,751	external/causalbench/dataset_k562_filtered.npz.json /cells_out *(excluded)*
cb.k562.genes	622	external/causalbench/dataset_k562_filtered.npz.json /genes_out *(excluded)*
cb.k562.controls	10,691	external/causalbench/dataset_k562_filtered.npz.json /controls *(excluded)*
cb.rpe1.cells	162,733	external/causalbench/dataset_rpe1_filtered.npz.json /cells_out *(excluded)*
cb.rpe1.genes	383	external/causalbench/dataset_rpe1_filtered.npz.json /genes_out *(excluded)*
cb.rpe1.controls	11,485	external/causalbench/dataset_rpe1_filtered.npz.json /controls *(excluded)*